Geological Maps – 1

Orientation of a plane: dip and strike

Beds (strata) originate as approximately horizonal layers that may subsequently become folded during tectonic events leading to the formation of mountain belts. Beds are geometrically considered as planes, and geologists record planar orientations by recording two pieces of information.

(1) Strike: is the direction of a horizontal line on a plane.
(2) Dip: is the maximum angle of slope of the plane and is always perpendicular to strike.

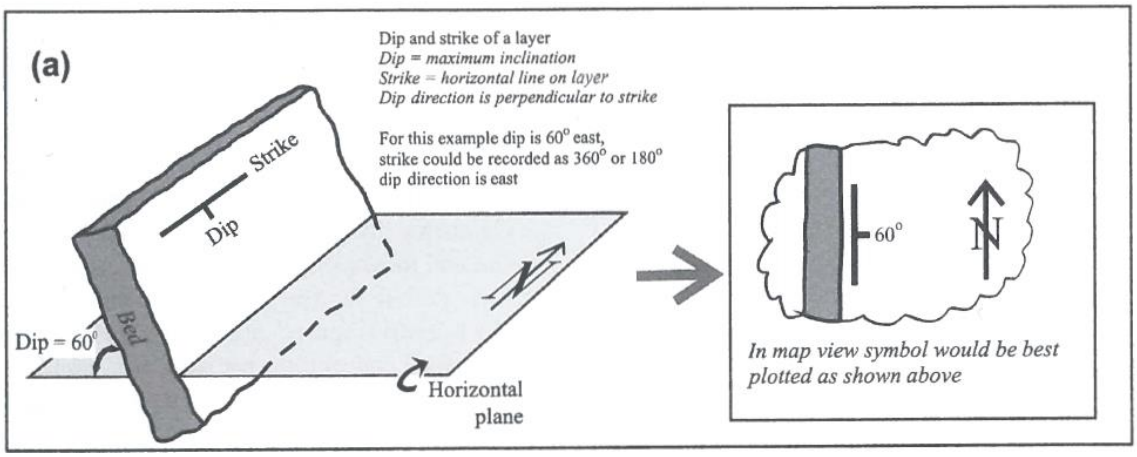


Figure 1 Explanation of the strike and dip measurements of geological surfaces.

Fig. 1 illustrates the strike and dip on an inclined plane and shows the strike/dip symbol that is often used on geological maps to indicate the orientation of strata.

Strike and dip readings are made in the field and are recorded using different formats, the one we will use is: **strike/dip** and **dip direction**. In the example shown in Fig 1 the information would be recorded as 180/60 E.

> **Exercise 1:** Look at the model of a dipping plane and identify the strike line and the maximum dip angle. Check the strike is horizontal using a spirit level (use an app on your phone).

Layered rocks and Topography

Most Earth scientists us geological maps where the geological information is superimposed on the topographic relief. It is useful to know the elevation of the land surface because strata will outcrop at different elevations depending on the dip of their contacts. Figs 2 and 3 illustrates the principles, showing how topographic relief is important in controlling the shape of outcrop pattern on the map. The block diagrams in Fig. 2, show a stream in a valley (note the topographic contours V upstream) and a geological layer (stratum) outcropping on the cross-sections of the block.

(a) Horizontal stratum - the outcrop perfectly follows the topographic contours.
(b) Vertical stratum – shows no sympathy with topography and cuts across the hillside
(c) Dipping stratum into the hill shows an outcrop that V's upstream and cuts across the topographic contours.

(d) Dipping stratum in a downstream direction more steeply than the slope of the stream forms a V downstream that cut across the topographic contours.

When a dipping contact crosses a valley, it forms a "V", with the apex pointing in the direction of dip of the contact (Fig. 2c-d). The main exceptions are where the dip of the contact is horizontal (Fig. 2a) or vertical (Fig. 2b).

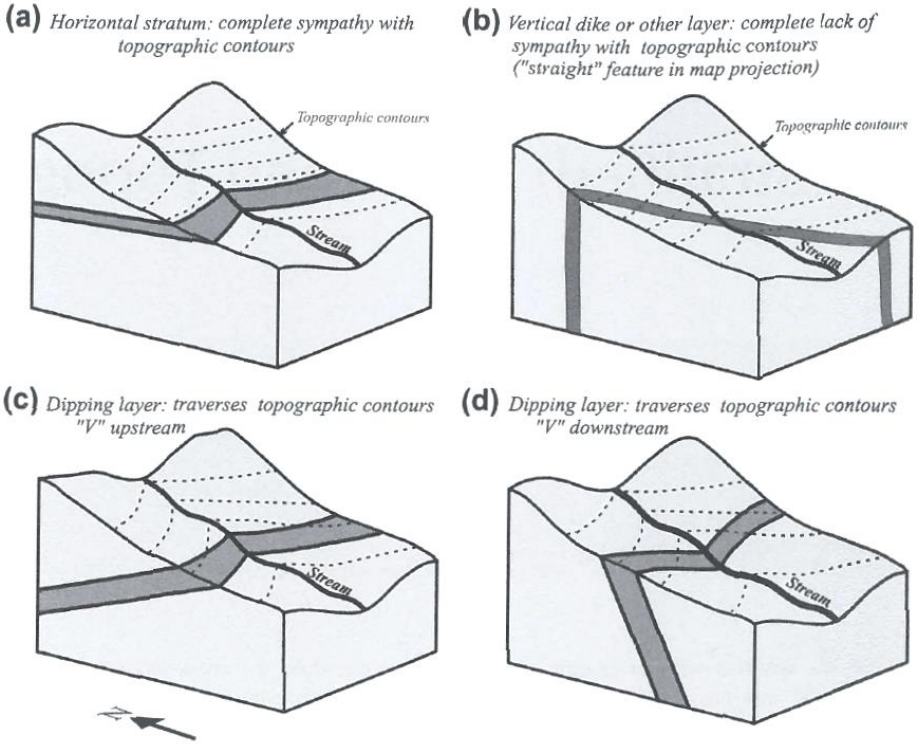


Figure 2. Block diagrams showing level of similarity between the shapes of outcrops boundaries and topography.

Fig. 3 shows how the outcrop pattern occurs across a hillside in map and cross-section views.

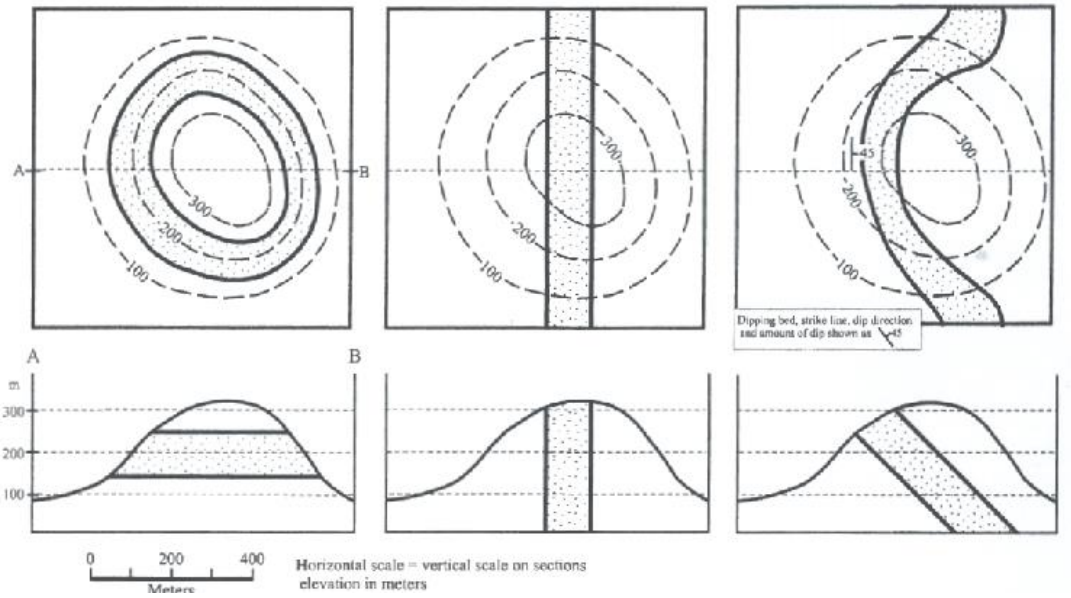


Figure 3. Topographic expressions of outcrop boundaries shown in map view.

Let's use the relationship between outcrop patterns and topography to make some simple interpretations using a real geological map shown in Fig 4.

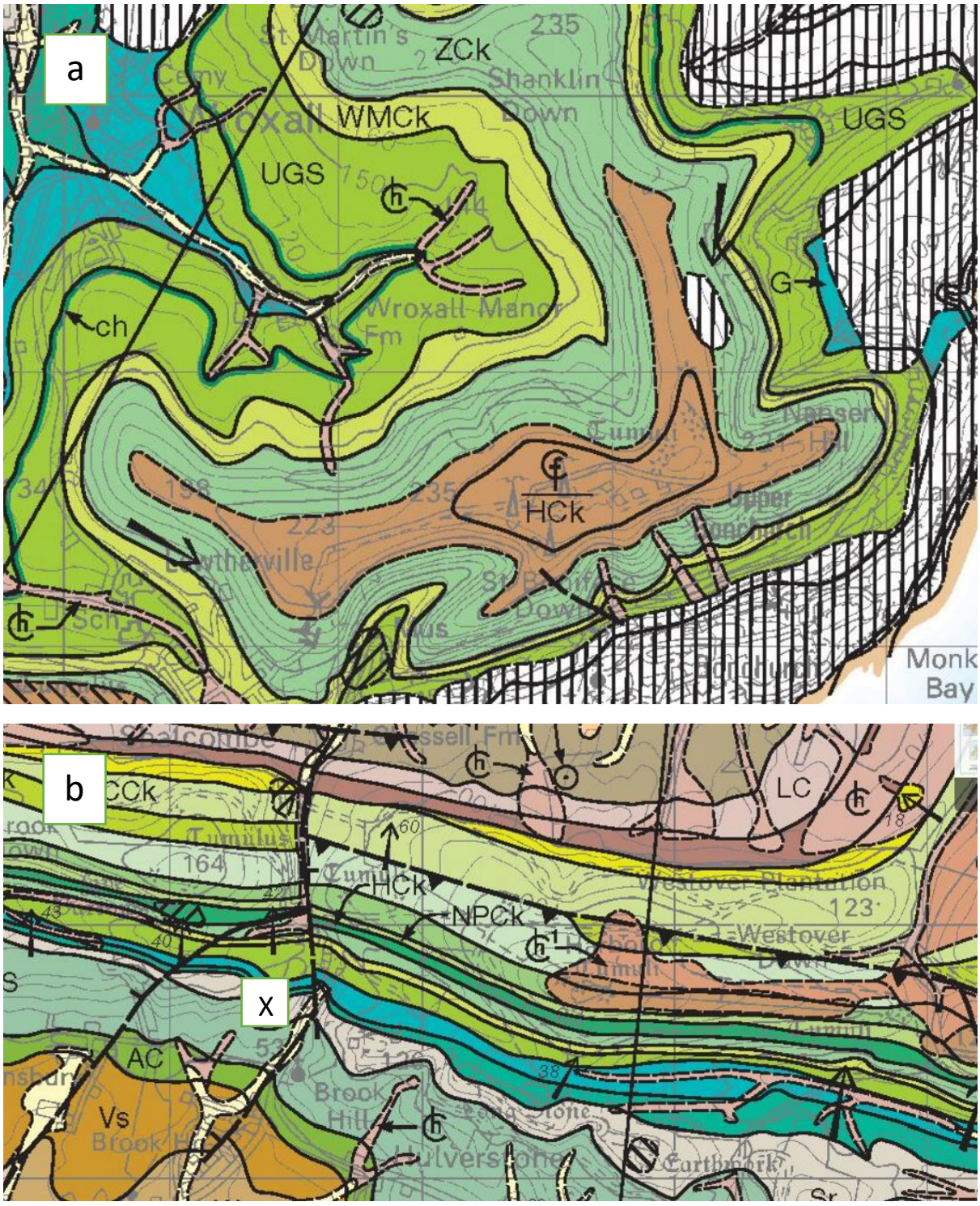


Figure 4 (a) and (b) show small areas from the geological map of the Isle of Wight. The grid squares are 1km and north is upwards, parallel to the easting grid lines.

Exercise 2: Two small areas of the geological map for the Isle of Wight are shown in Figure 4.

(1) In Fig, 4(a), observe the interaction between the geological contacts and topographic contours. What do you infer about the dip of the beds in this area?

(2) Compare this with Fig. 4(b), what do you infer about the dip relative to (a)?

(3) Look at the feature at 'X' in (b), based on what you observe what direction is the dip of the beds in this area?

If you would like to see more detail of this geological map The British geological Survey have produced an on-line map viewer for many areas of the UK. The link to the Isle of Wight map is here:

https://largeimages.bgs.ac.uk/iip/mapsportal.html?id=1004974

A simple construction method for mapping geological contacts of strata

How does a geologist, using information from a few scattered exposures, determine the direction of bedding contacts across hillsides and valleys? Often a reliable geological map needs to be made of an area where perhaps only a few % of the land surface is exposed bedrock.

If we assume in the simplest case of beds being represented by plane-dipping layers then we can use a method sometimes known as a **three-point problem** to extend bedding contacts to complete a map. This relies on having three locations in an area where the same bedding contact is exposed and involves the construction of **stratum contours**.

Stratum contours are the points of intersection of a geological plane and topographic contours. For a uniformly dipping plane, stratum contours are straight, parallel lines, equally space corresponding to higher and lower elevations of a plane (Fig. 5).

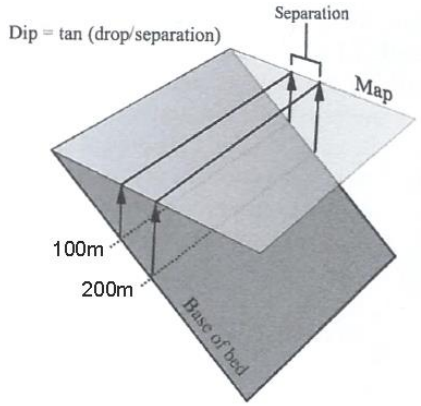
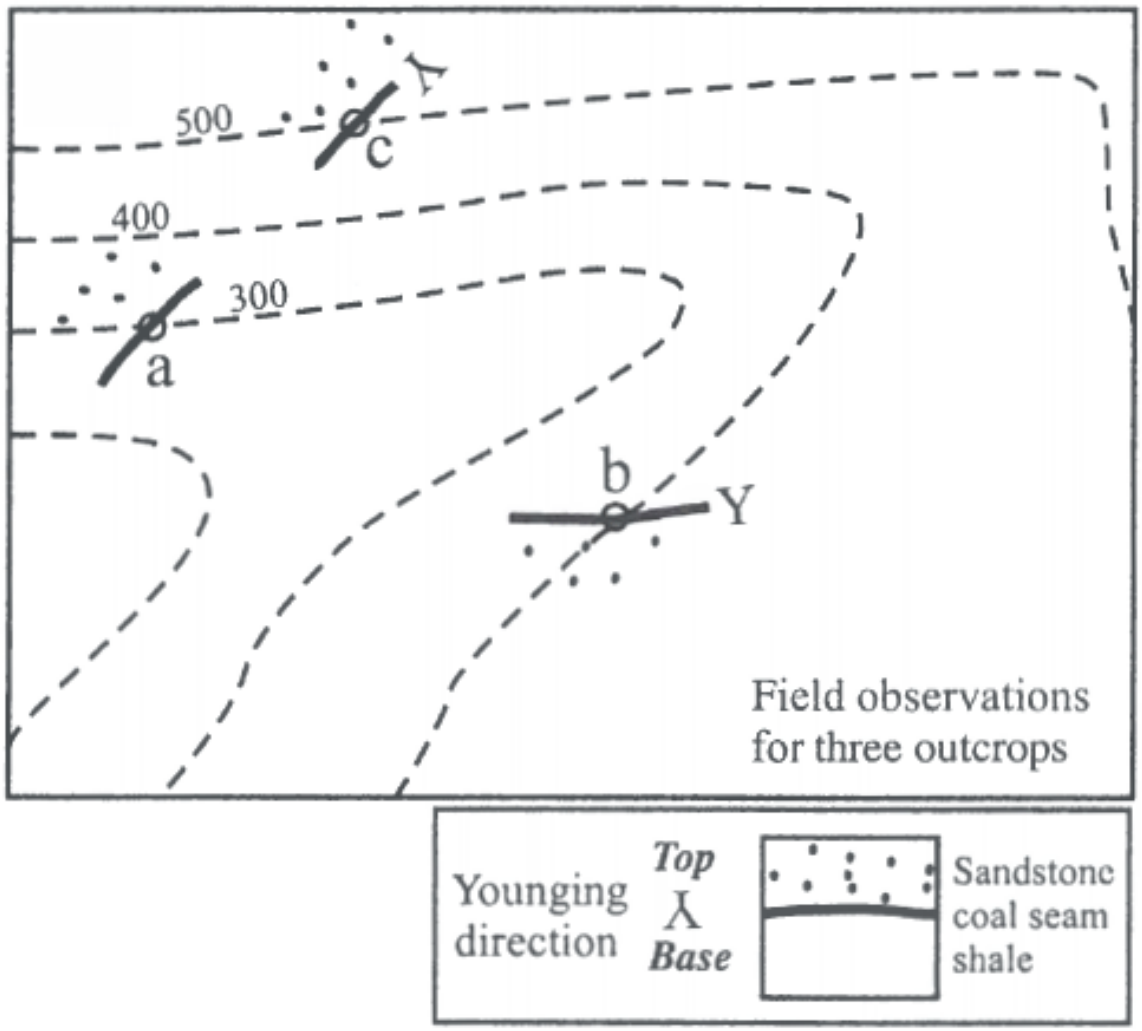


Figure 5 A plane-dipping layer extending below the land surface at an angle given by the dip. Stratum contours are marked on the plane for two different depths below the surface.

Exercise 3. The map below shows the outcrop of a thin coal seam (the solid black line), with younger age sandstone above and an older shale below (note the use of younging symbols on the map). The dashed lines are topographic contours in metres and the characteristic up-elevation V shape formed by the contours indicates the coal seam is crossing a valley.

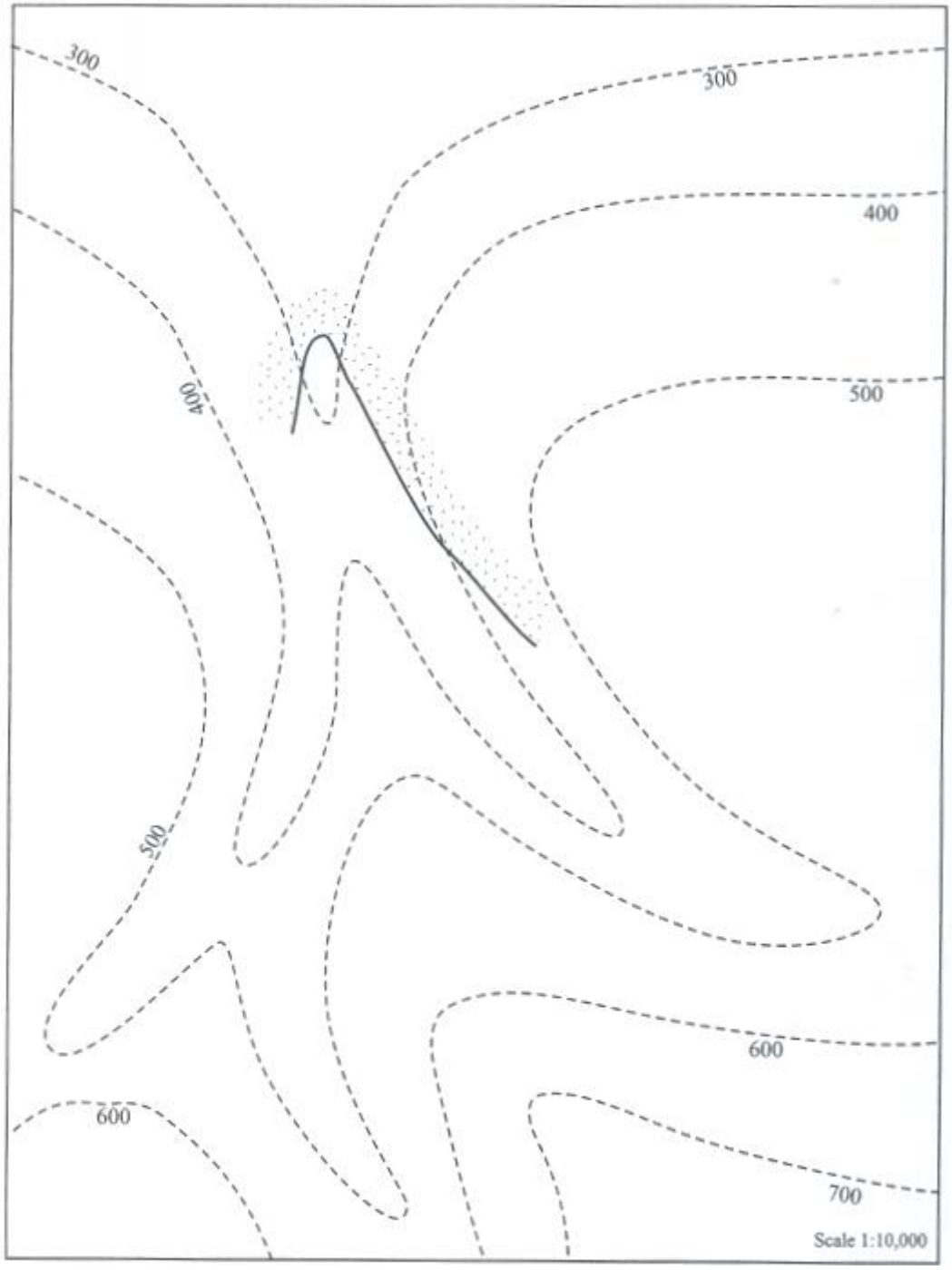


1. Complete the map of plane-dipping beds (three point problem). You will be shown how to do this.

2. The interaction between the outcrop pattern and the topographic contours can be used to inform on the direction of dip of the coal seam. What is the inferred direction of dip?

3. In addition to extrapolating geological contacts structure contours can be used to calculate the strike and dip of beds. Assume the map has a scale of 1:4,000. Calculate the angle of dip and measure the strike. Record your answer as strike/dip and dip direction of the coal seam.

Exercise 4: Complete the map (three point problem). Indicate the strike and dip of the beds and show which bed is older.



Many of the diagrams used in this practical are from Borradaile G. (2014) Understanding Geology Through Maps. First Edition, Elsevier, 183 pp.